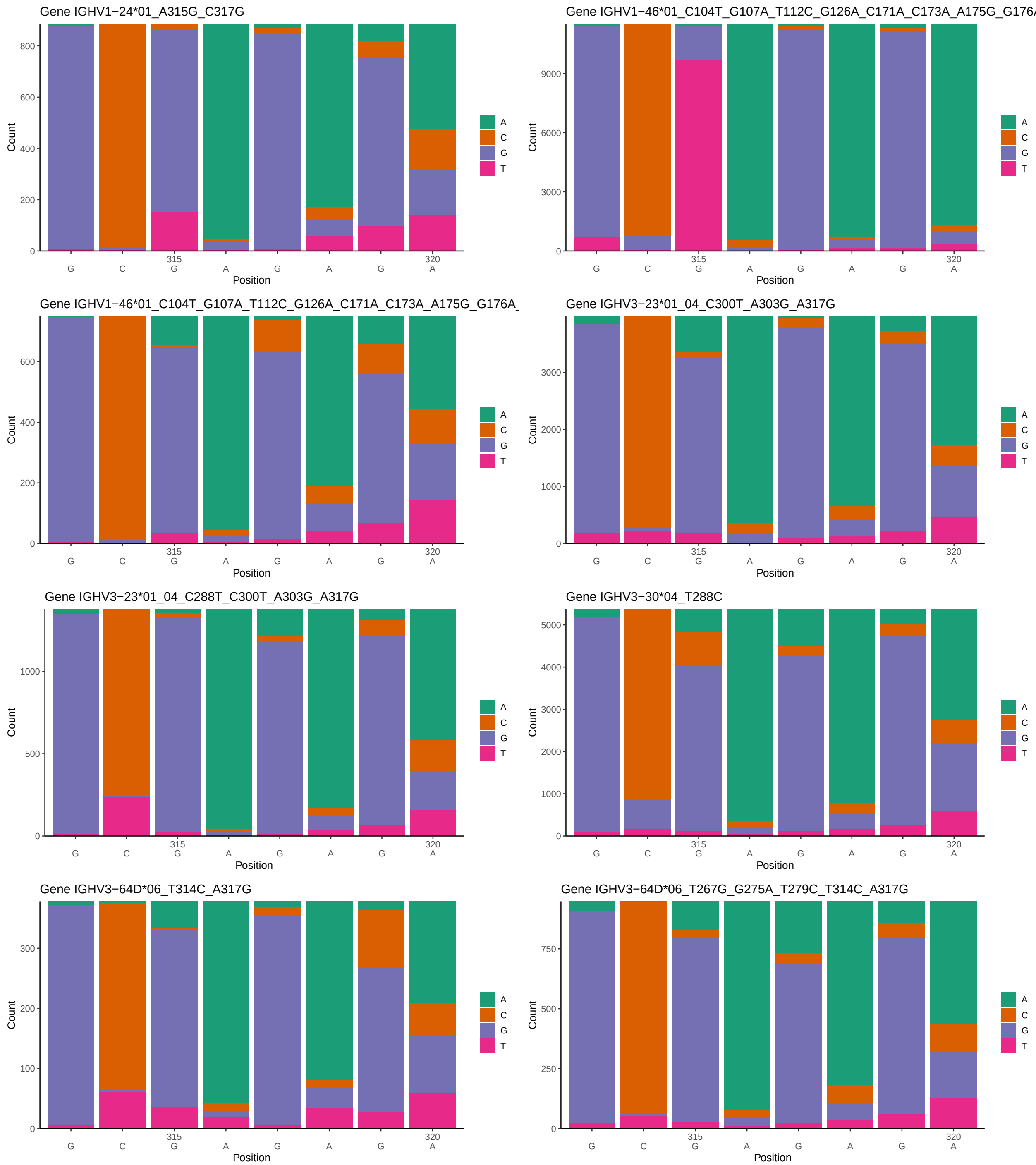
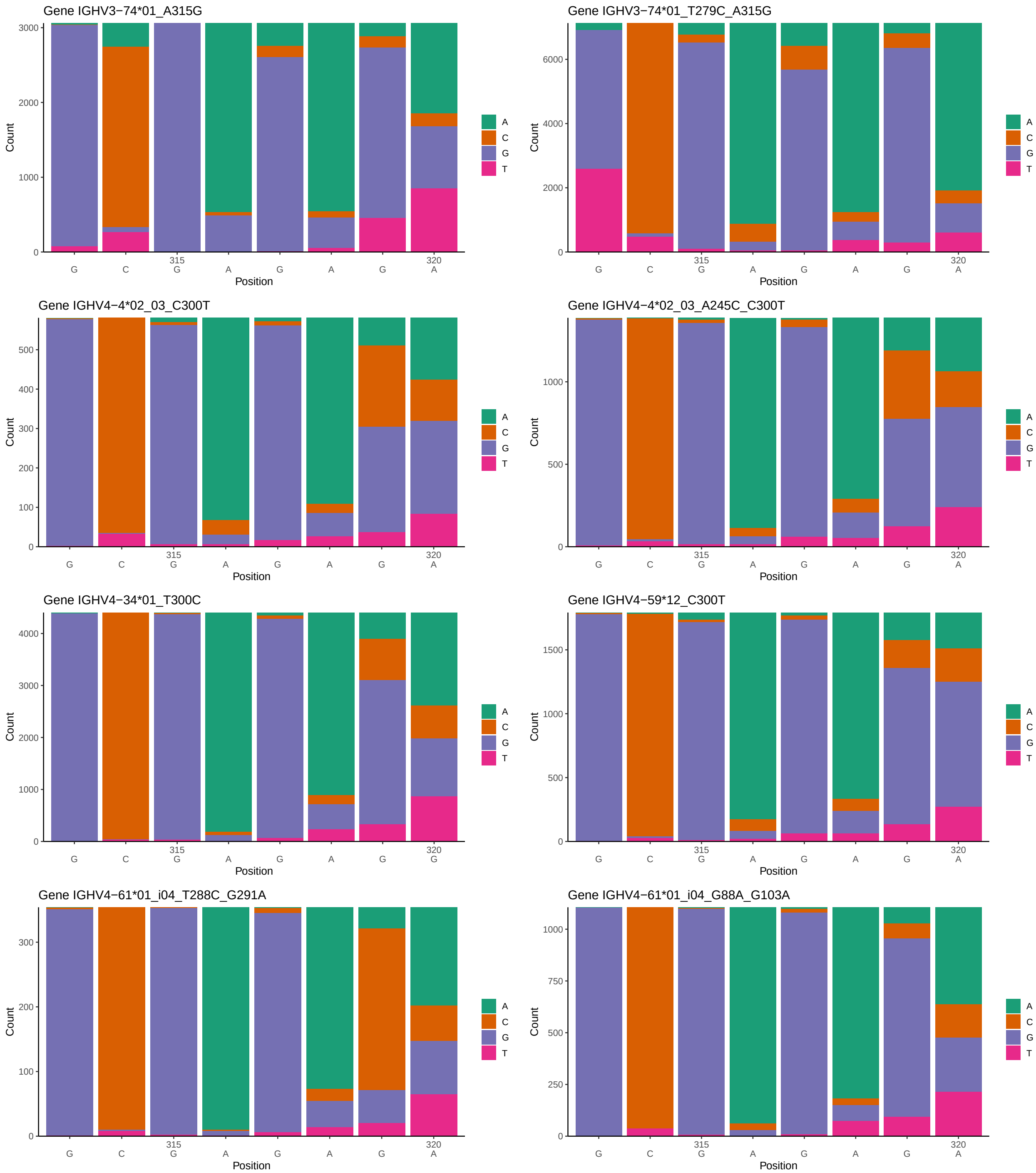
OGRDBstats Report

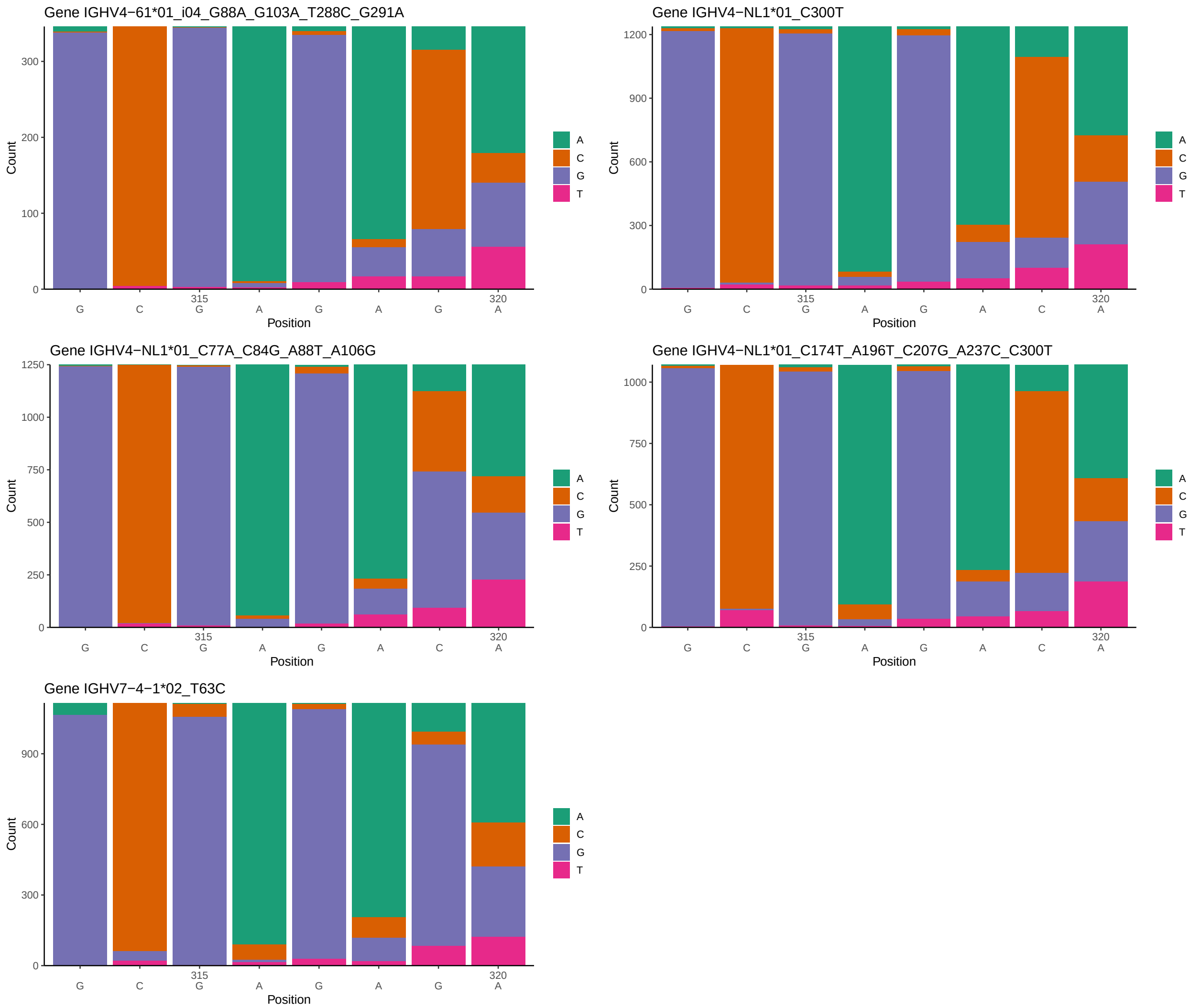
Contents

1 Novel sequence analysis

1.1 End-nucleotide composition

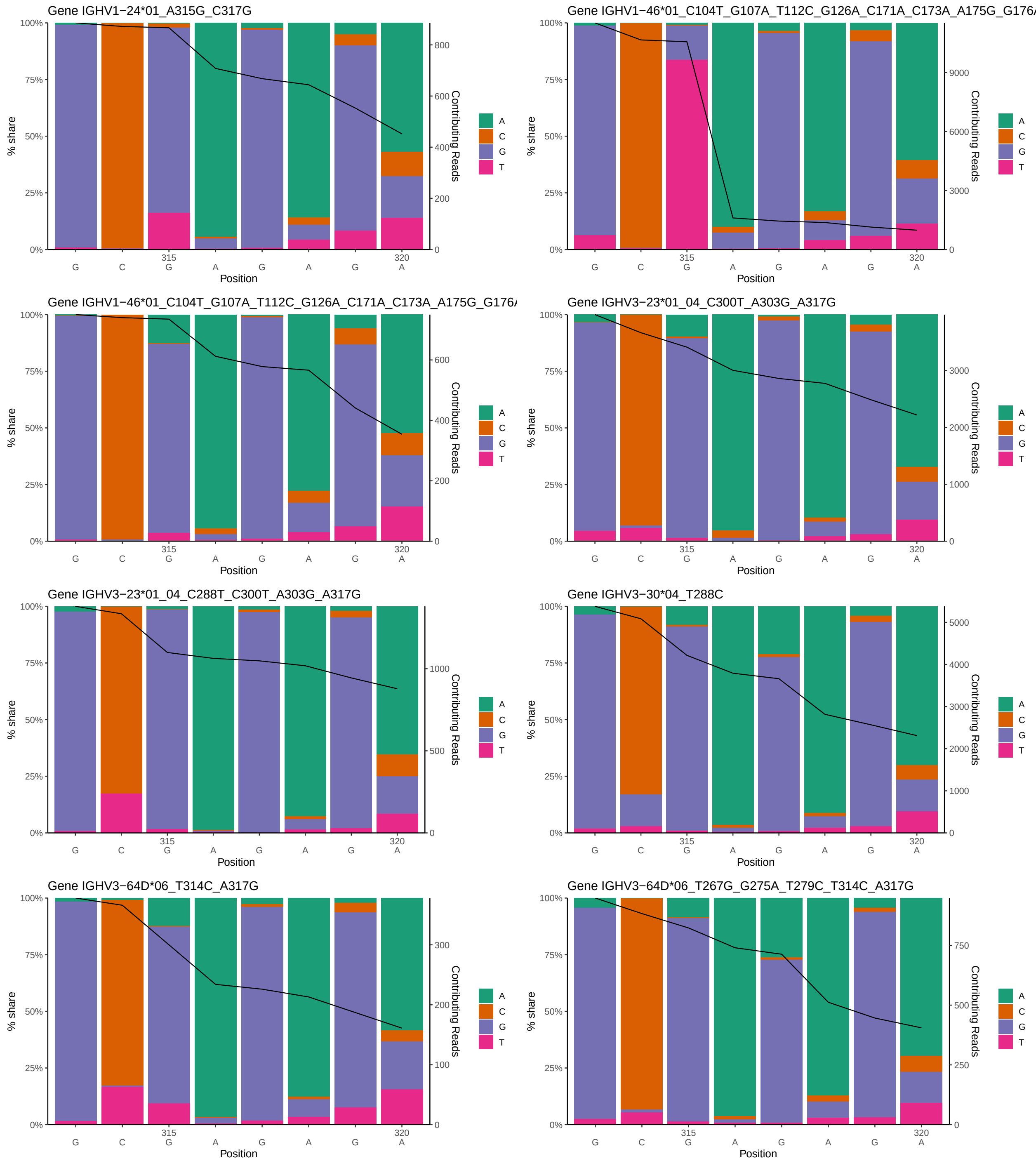


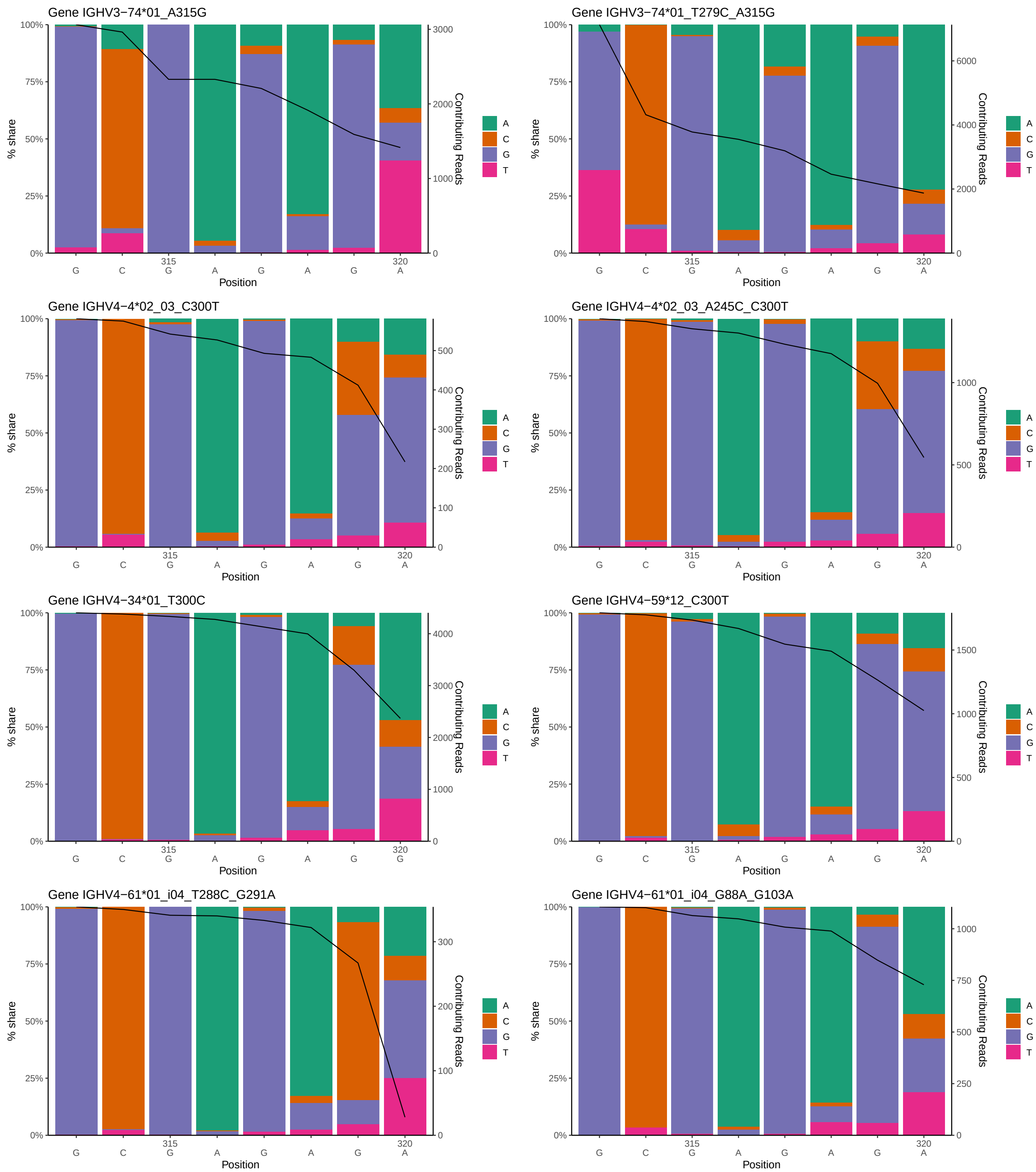

Gene IGHV3−74*01_A315G
Gene IGHV3−74*01_T279C_A315G
Gene IGHV4−4*02_03_C300T
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−34*01_T300C
Gene IGHV4−59*12_C300T
Gene IGHV4−61*01_i04_T288C_G291A
Gene IGHV4−61*01_i04_G88A_G103A
Count
Position
A
C
G
T



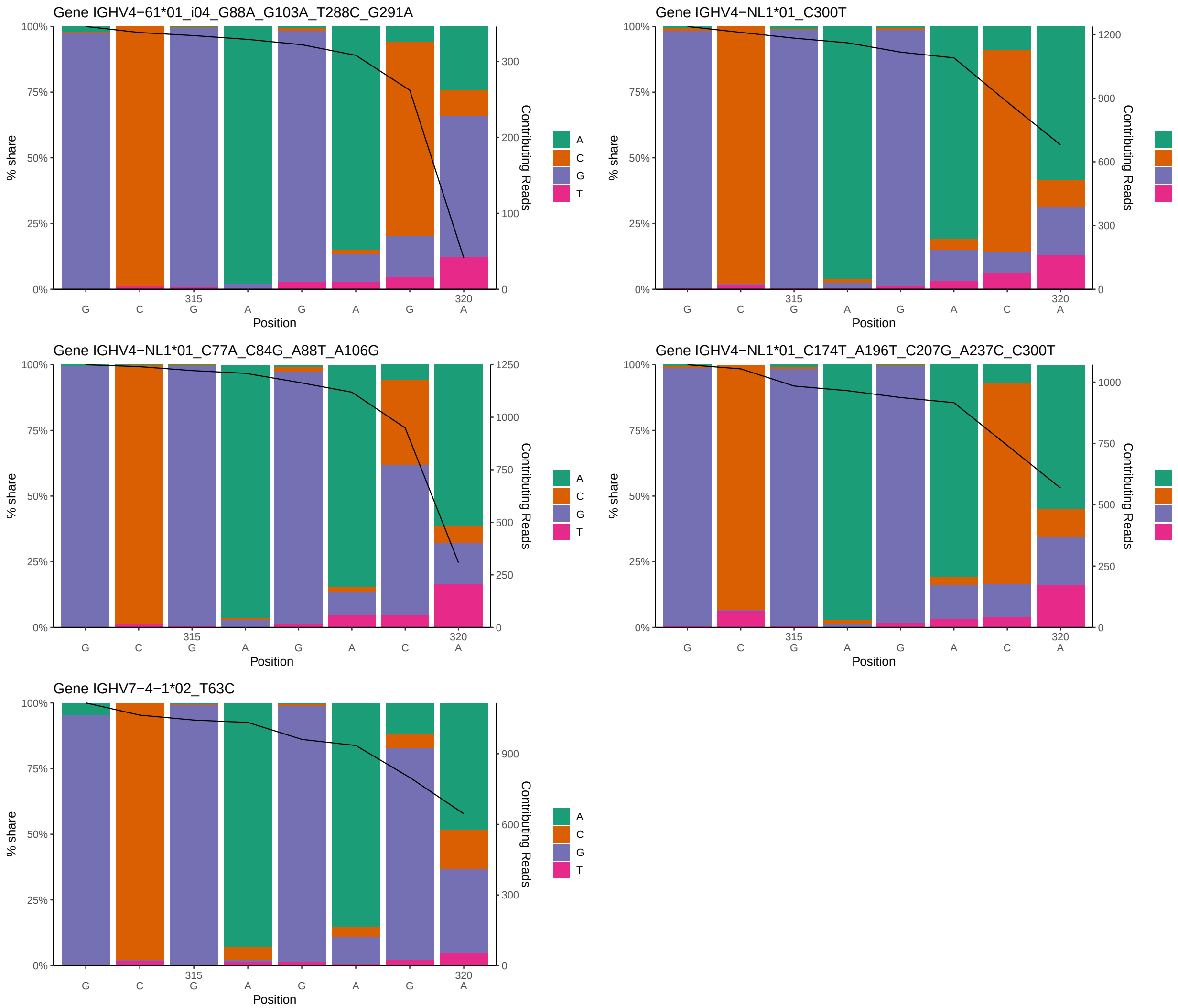
Gene IGHV4−61*01_i04_G88A_G103A_T288C_G291A
Count
0
100
200
300
G
C
315
G
A
G
A
G
320
A
Position
A
C
G
T
Gene IGHV4−NL1*01_C300T
Count
0
300
600
900
1200
G
C
315
G
A
G
A
C
320
A
Position
A
C
G
T
Gene IGHV4−NL1*01_C77A_C84G_A88T_A106G
Count
0
250
500
750
1000
1250
G
C
315
G
A
G
A
C
320
A
Position
A
C
G
T
Gene IGHV4−NL1*01_C174T_A196T_C207G_A237C_C300T
Count
0
250
500
750
1000
G
C
315
G
A
G
A
C
320
A
Position
A
C
G
T
Gene IGHV7−4−1*02_T63C
Count
0
300
600
900
G
C
315
G
A
G
A
G
320
A
Position
A
C
G
T

1.2 Per-nucleotide consensus where previous nucleotides match the consensus

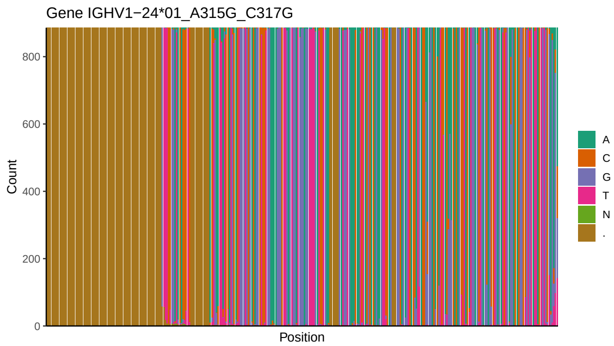


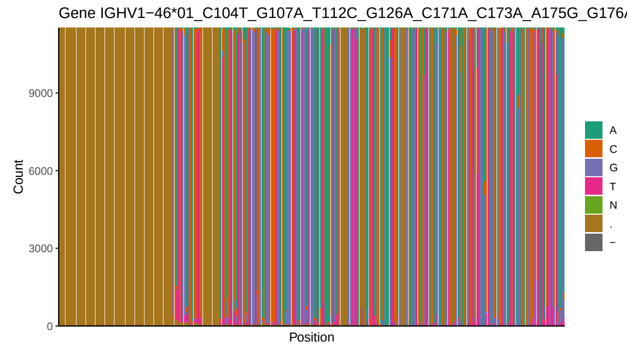


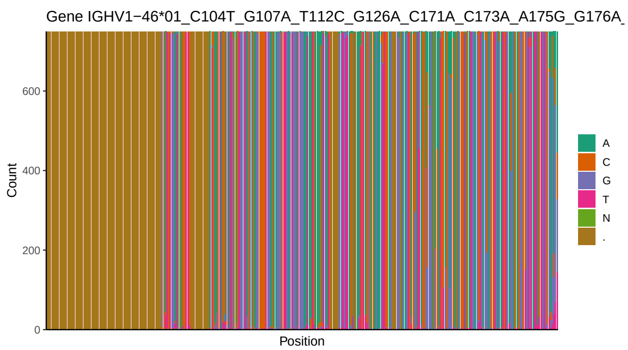
Gene IGHV3−74*01_A315G
Gene IGHV3−74*01_T279C_A315G
Gene IGHV4−4*02_03_C300T
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−34*01_T300C
Gene IGHV4−59*12_C300T
Gene IGHV4−61*01_i04_T288C_G291A
Gene IGHV4−61*01_i04_G88A_G103A
% share
Contributing Reads
Position
A
C
G
T

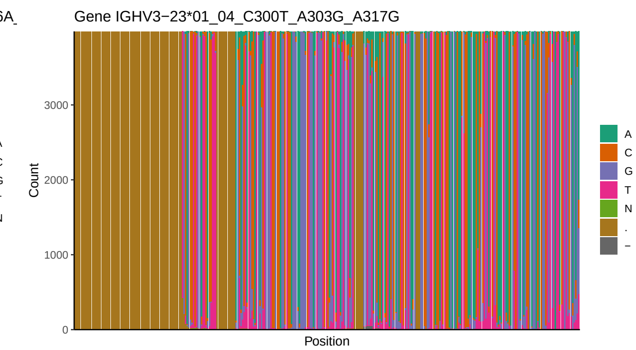

Gene IGHV4−61*01_i04_G88A_G103A_T288C_G291A
Gene IGHV4−NL1*01_C300T
Gene IGHV4−NL1*01_C77A_C84G_A88T_A106G
Gene IGHV4−NL1*01_C174T_A196T_C207G_A237C_C300T
Gene IGHV7−4−1*02_T63C
% share
Contributing Reads
Position
A
C
G
T

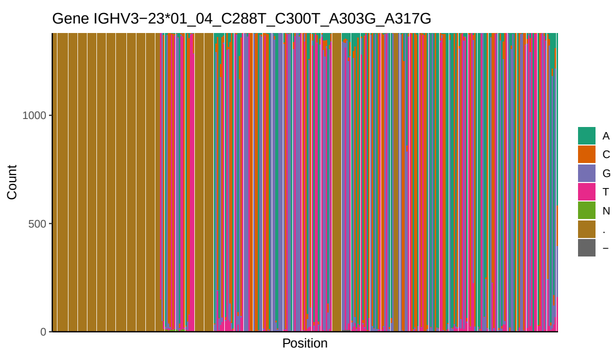
1.3 Whole-sequence composition of each assigned read

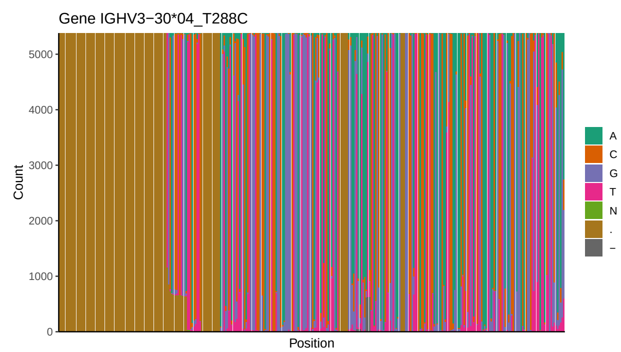


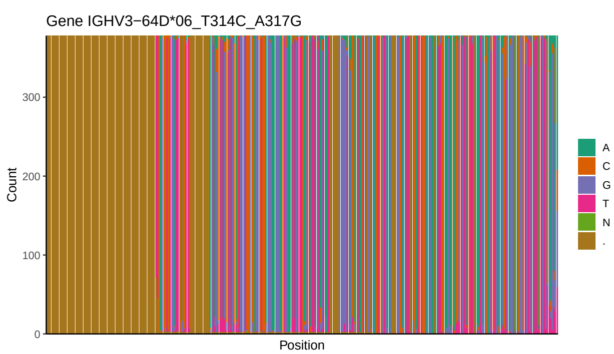


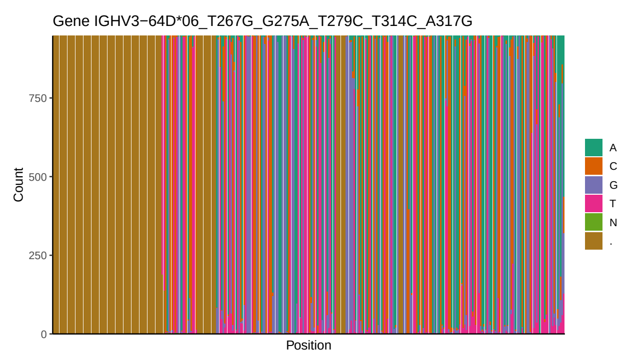


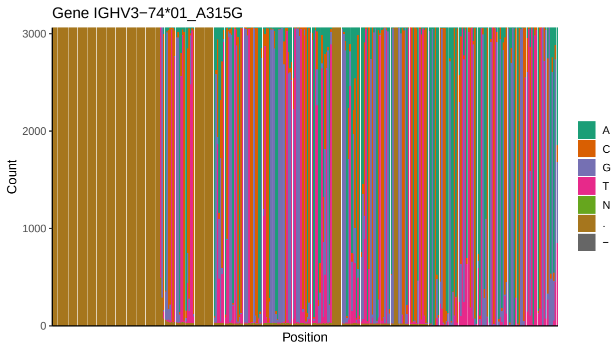


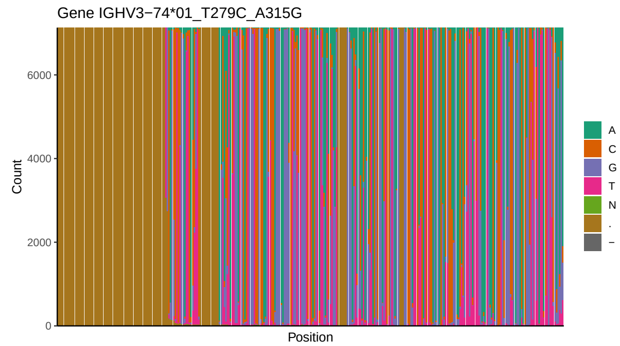


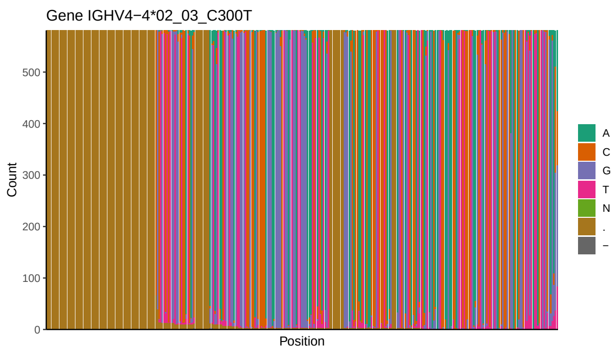


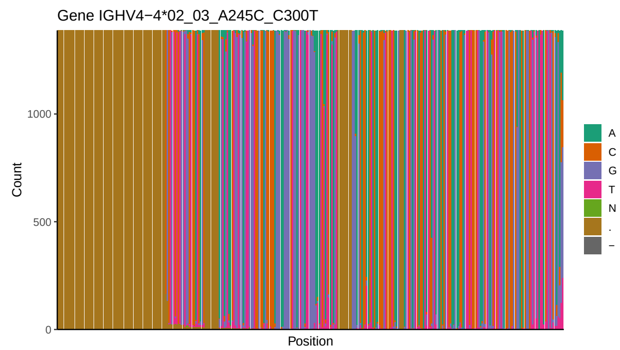


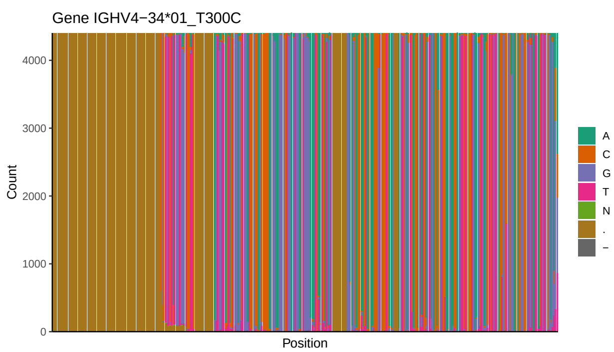


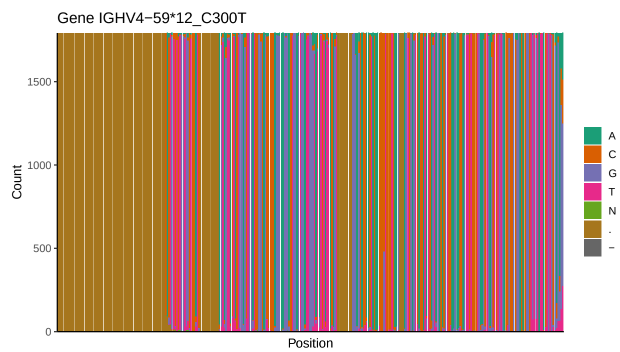

Gene IGHV3−74*01_A315G
3000
2000
1000
0
Count
Position
A
C
G
T
N
.
−

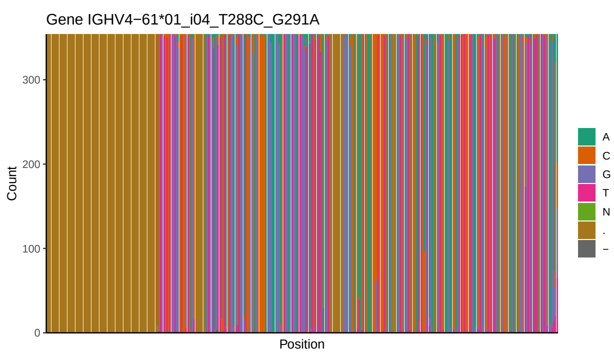

Gene IGHV3−74*01_T279C_A315G
6000
4000
2000
0
Count
Position
A
C
G
T
N
.
−

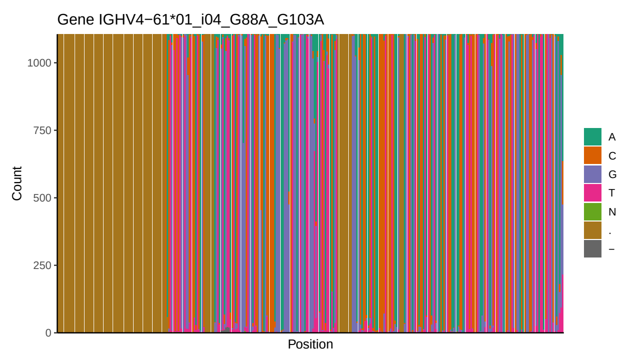

Gene IGHV4−4*02_03_C300T
500
400
300
200
100
0
Count
Position
A
C
G
T
N
.
−

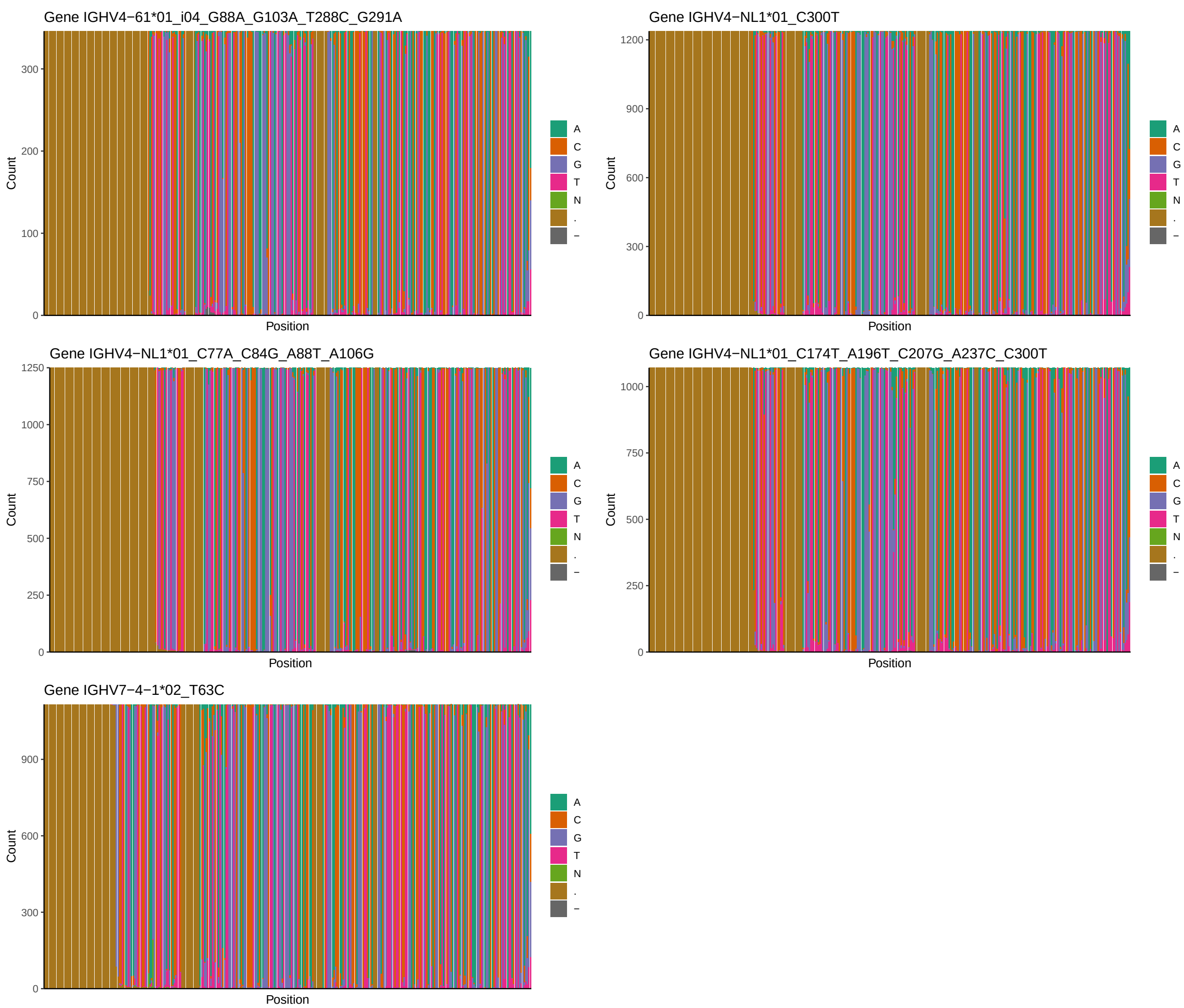

Gene IGHV4−4*02_03_A245C_C300T
1000
500
0
Count
Position
A
C
G
T
N
.
−


Gene IGHV4−34*01_T300C
4000
3000
2000
1000
0
Count
Position
A
C
G
T
N
.
−

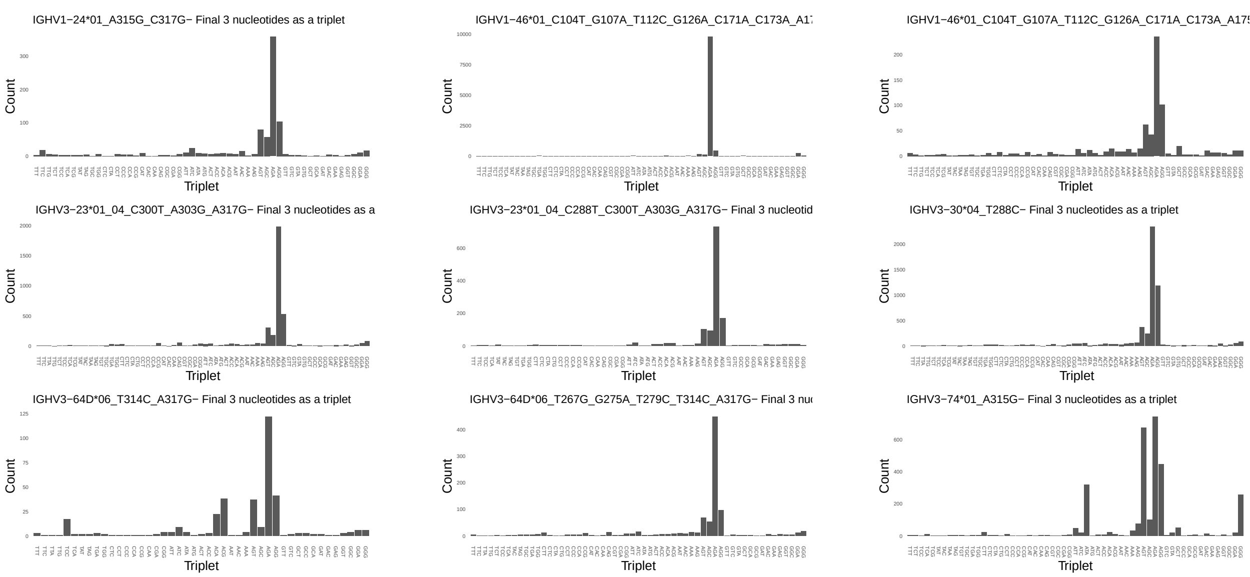

Gene IGHV4−59*12_C300T
1500
1000
500
0
Count
Position
A
C
G
T
N
.
−

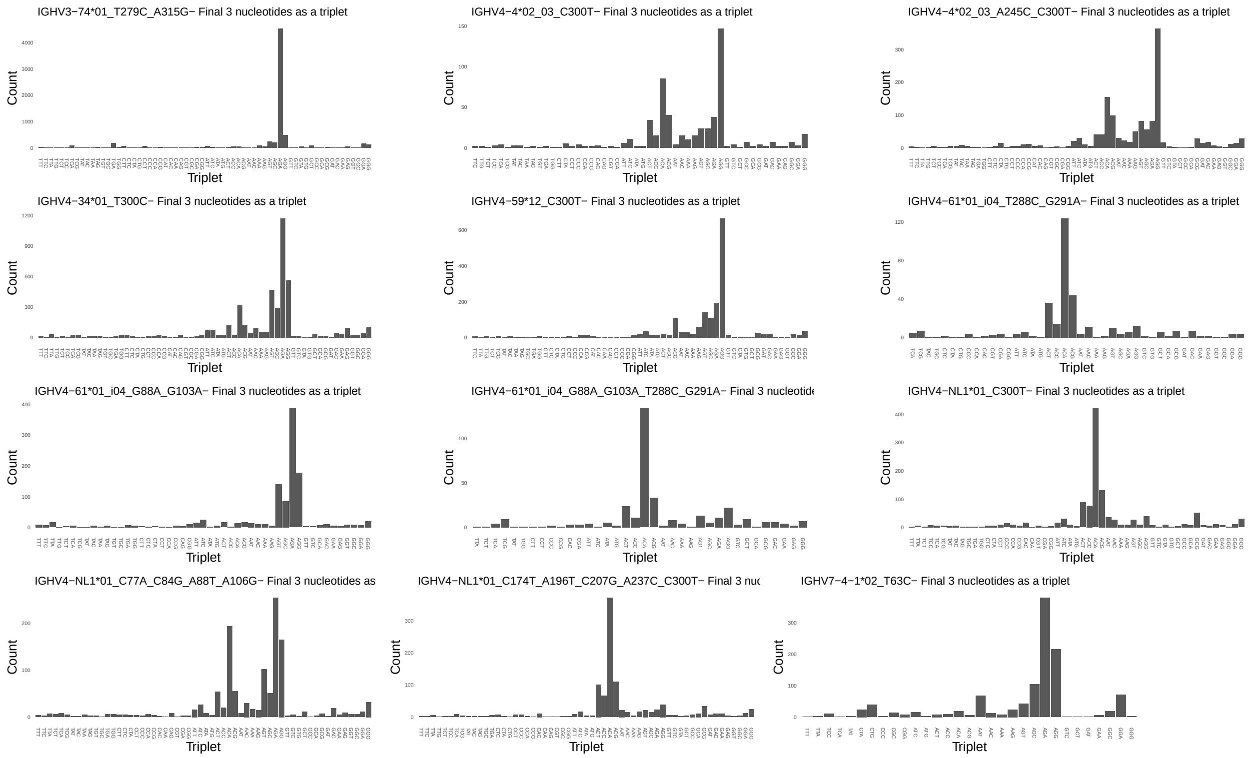

Gene IGHV4−61*01_i04_T288C_G291A
300
200
100
0
Count
Position
A
C
G
T
N
.
−


Gene IGHV4−61*01_i04_G88A_G103A
1000
750
500
250
0
Count
Position
A
C
G
T
N
.
−

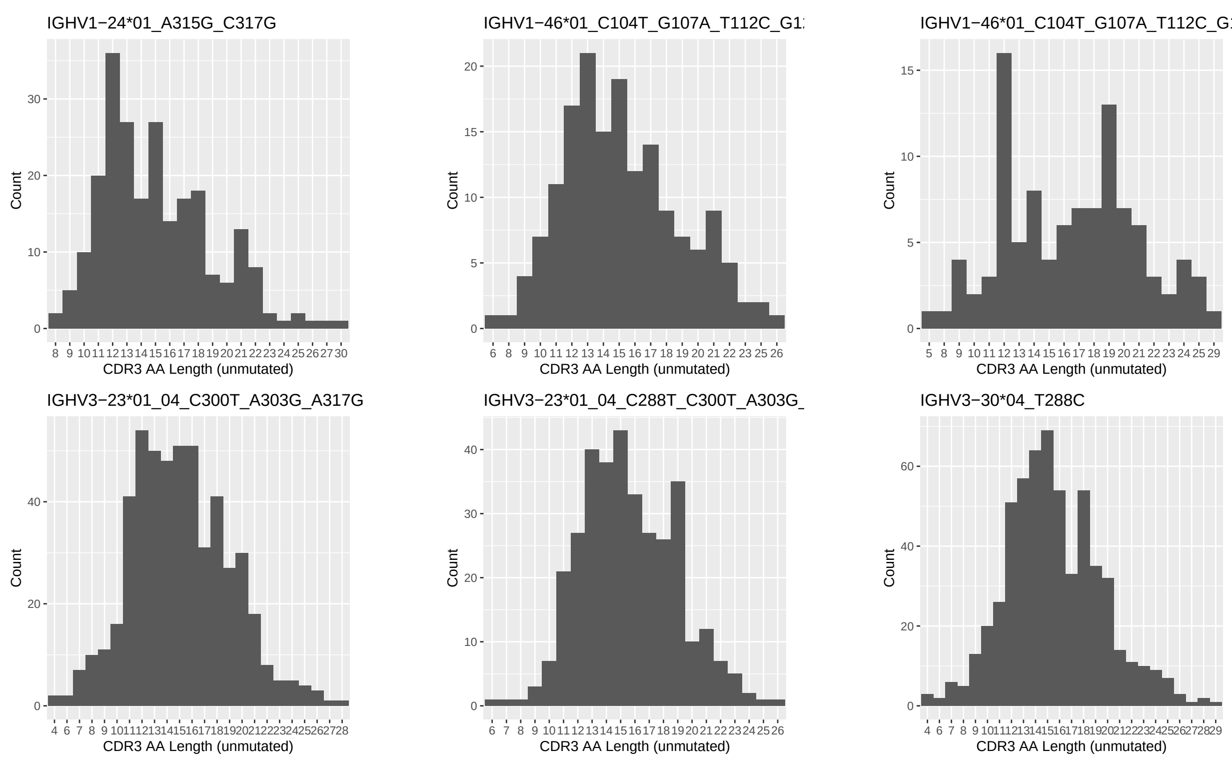


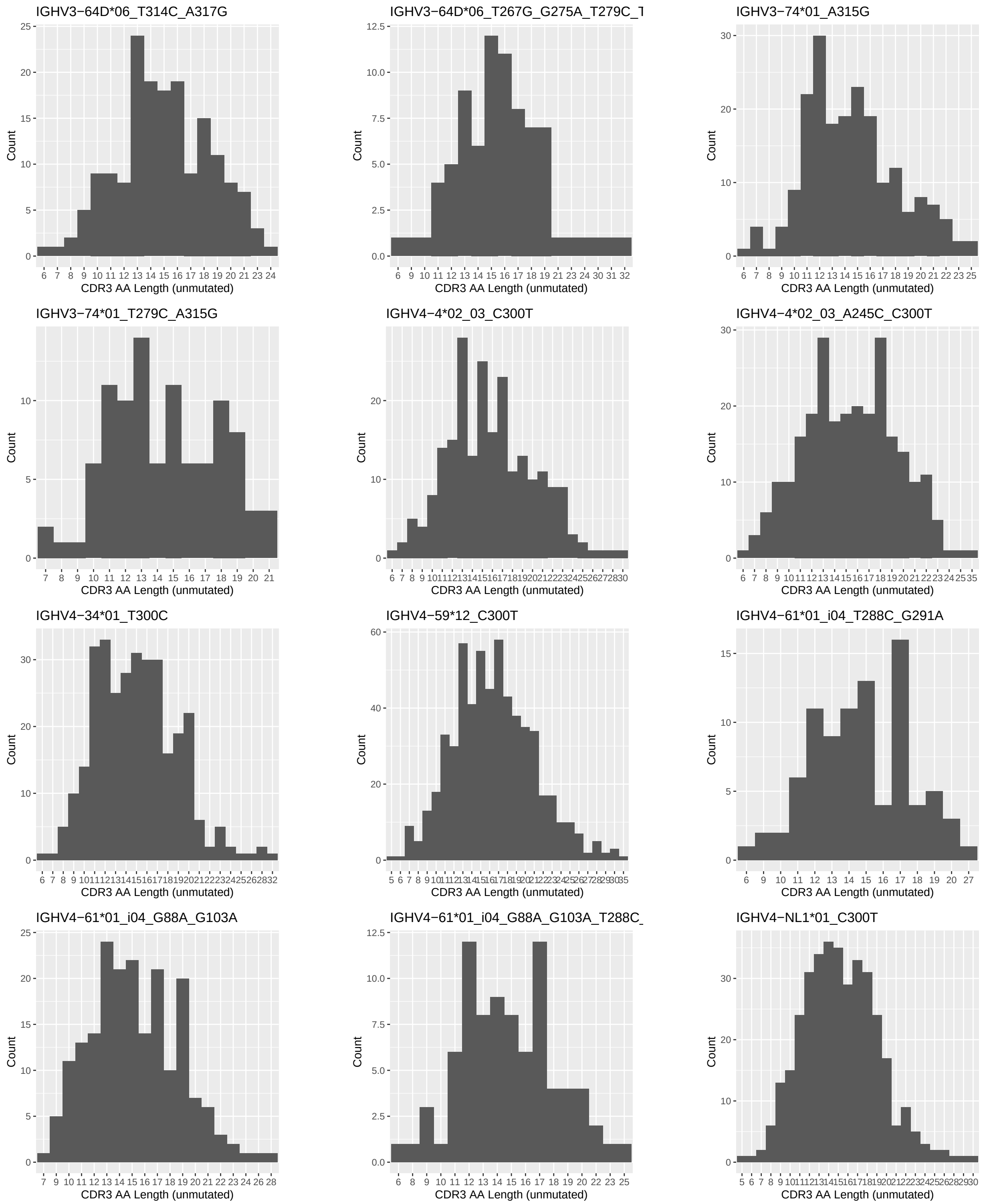
1.4 Final three nucleotides: frequency of each observed triplet

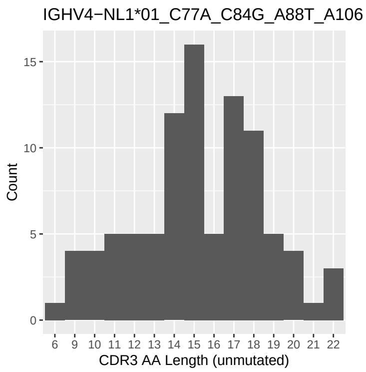


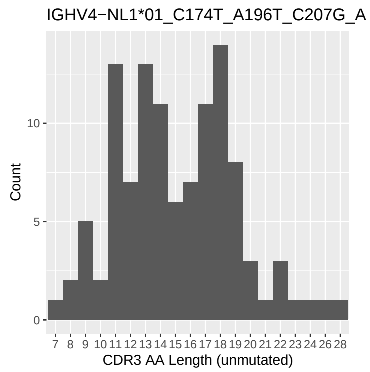


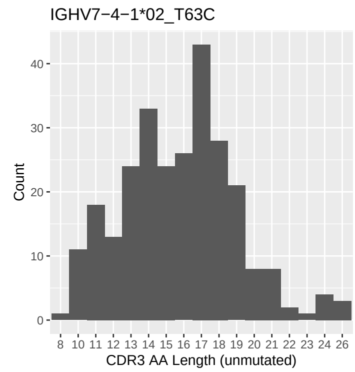
1.5 CDR3 length distribution, in assignments to novel alleles



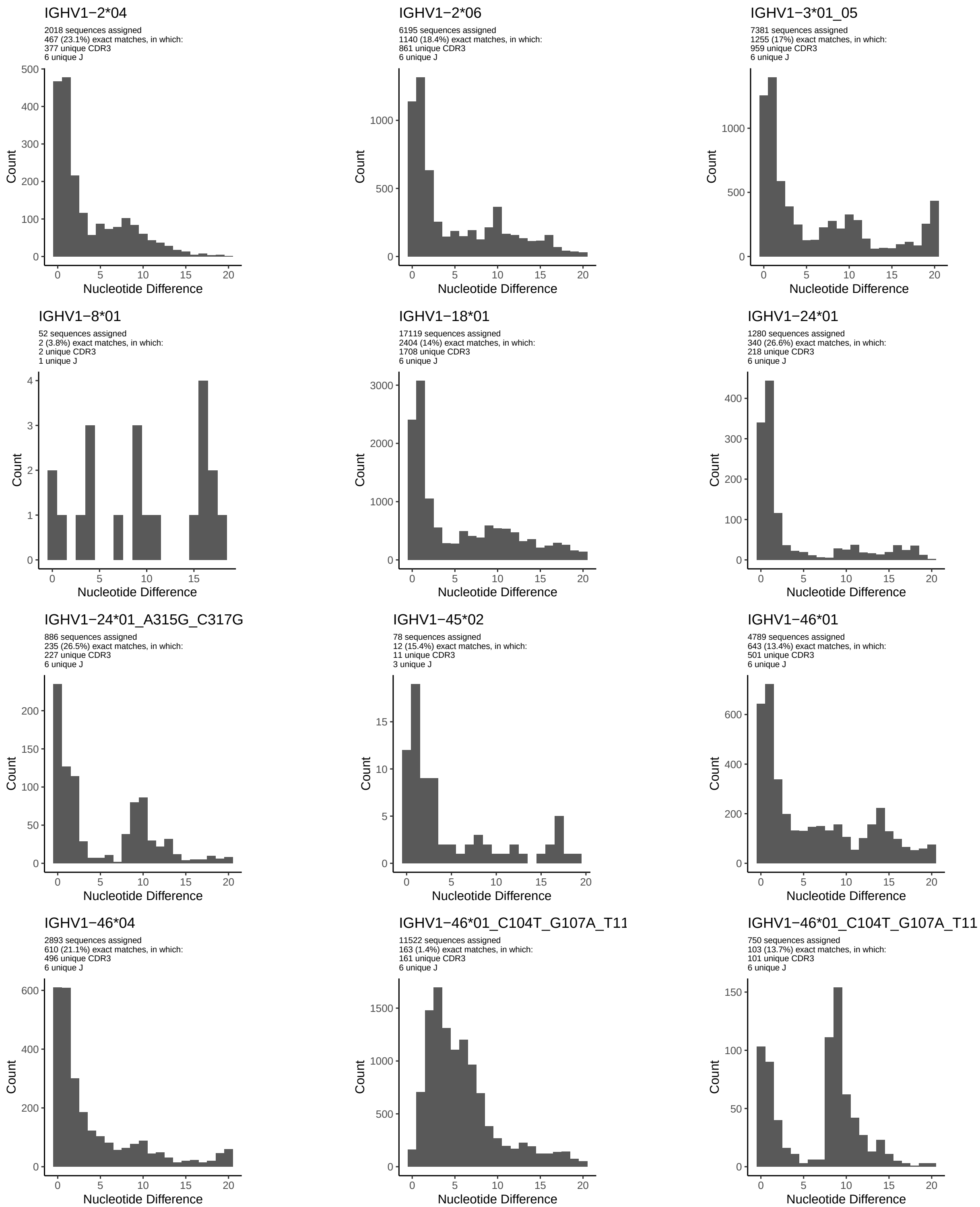

IGHV3−64D*06_T314C_A317G
IGHV3−64D*06_T267G_G275A_T279C_T
IGHV3−74*01_A315G
IGHV3−74*01_T279C_A315G
IGHV4−4*02_03_C300T
IGHV4−4*02_03_A245C_C300T
IGHV4−34*01_T300C
IGHV4−59*12_C300T
IGHV4−61*01_i04_T288C_G291A
IGHV4−61*01_i04_G88A_G103A
IGHV4−61*01_i04_G88A_G103A_T288C_
IGHV4−NL1*01_C300T
Count
CDR3 AA Length (unmutated)

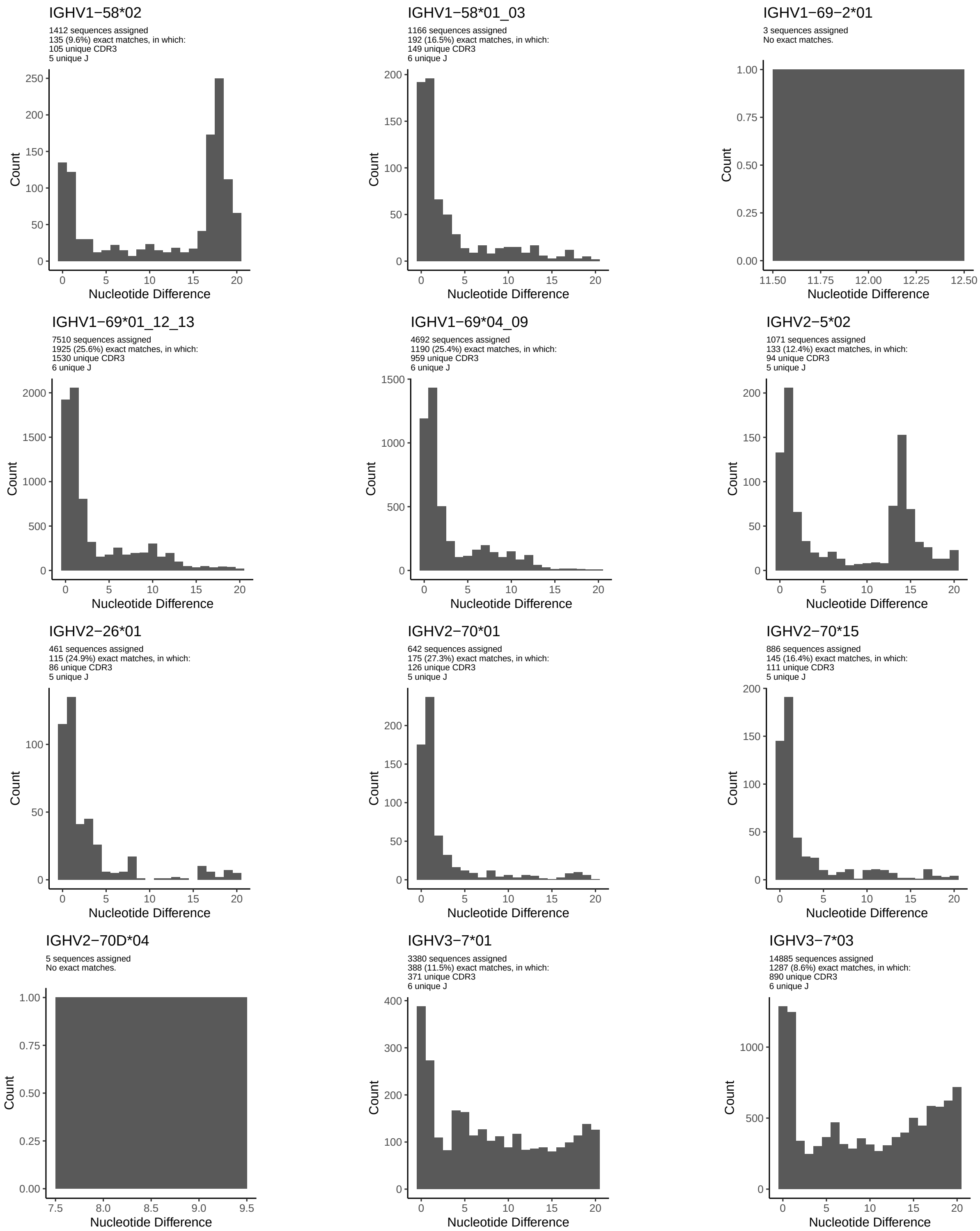

IGHV4−NL1*01_C77A_C84G_A88T_A106
Count
CDR3 AA Length (unmutated)


IGHV4−NL1*01_C174T_A196T_C207G_A
Count
CDR3 AA Length (unmutated)

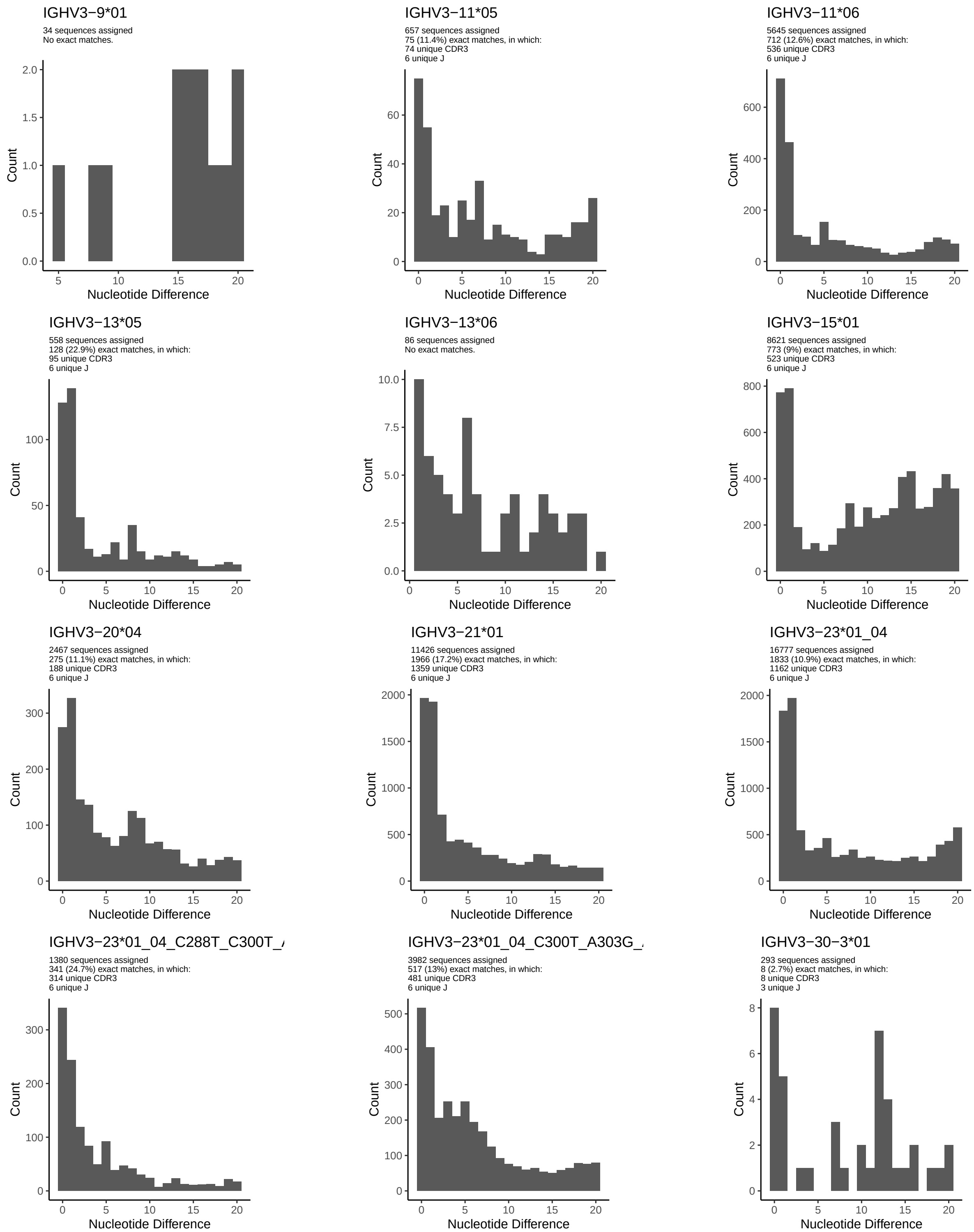

IGHV7−4−1*02_T63C
Count
CDR3 AA Length (unmutated)

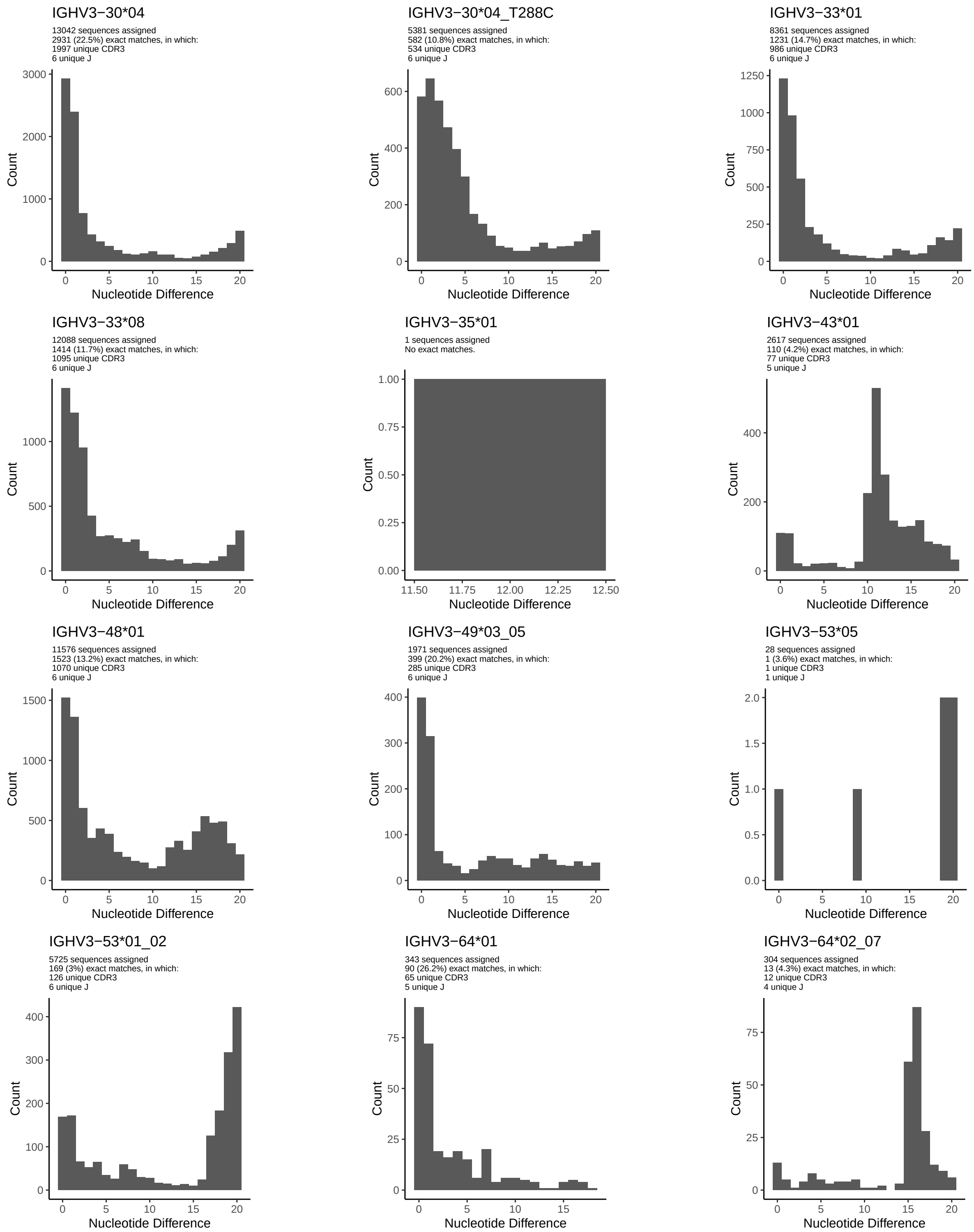
2 Variation from germline, in assignments to each allele

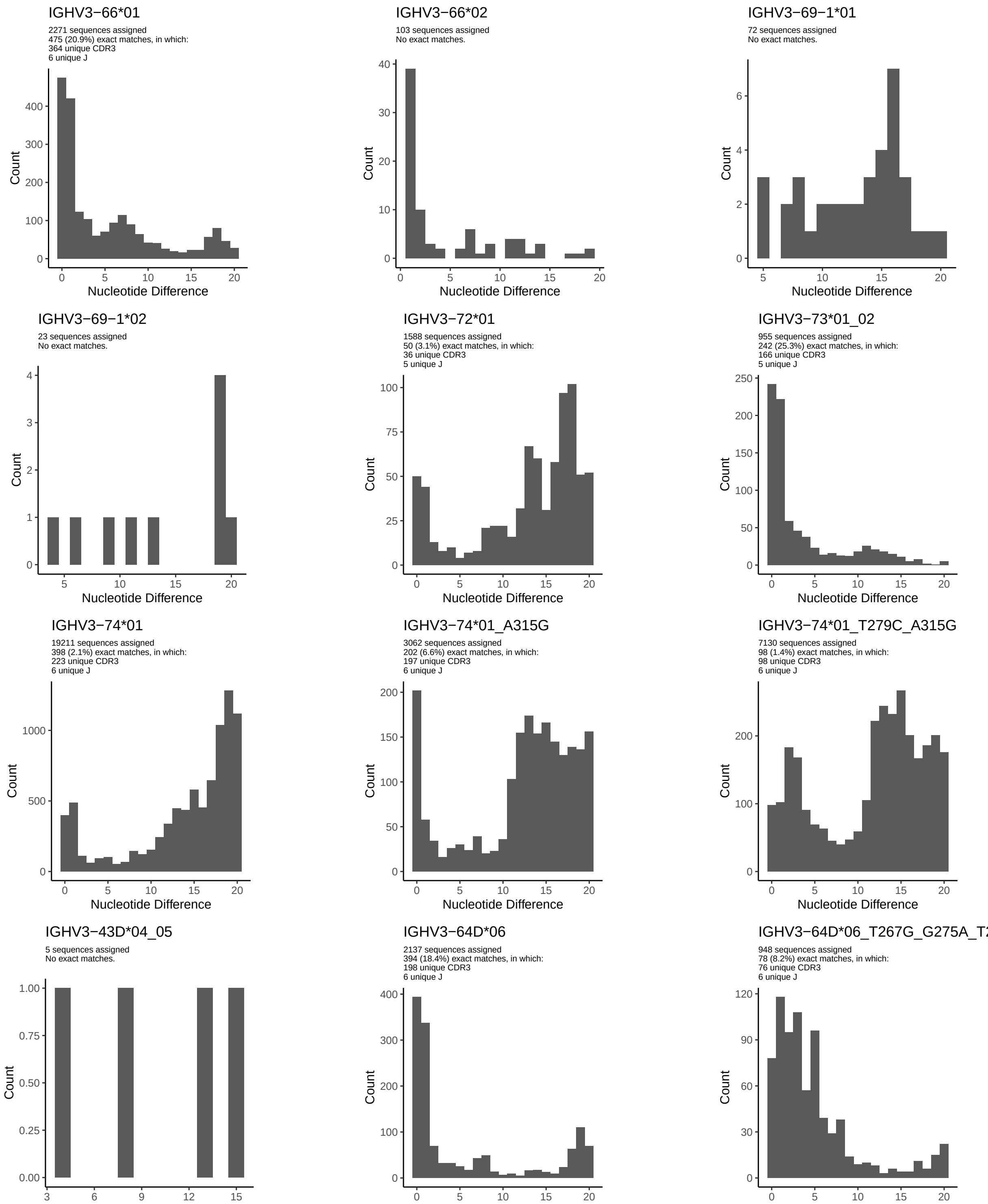


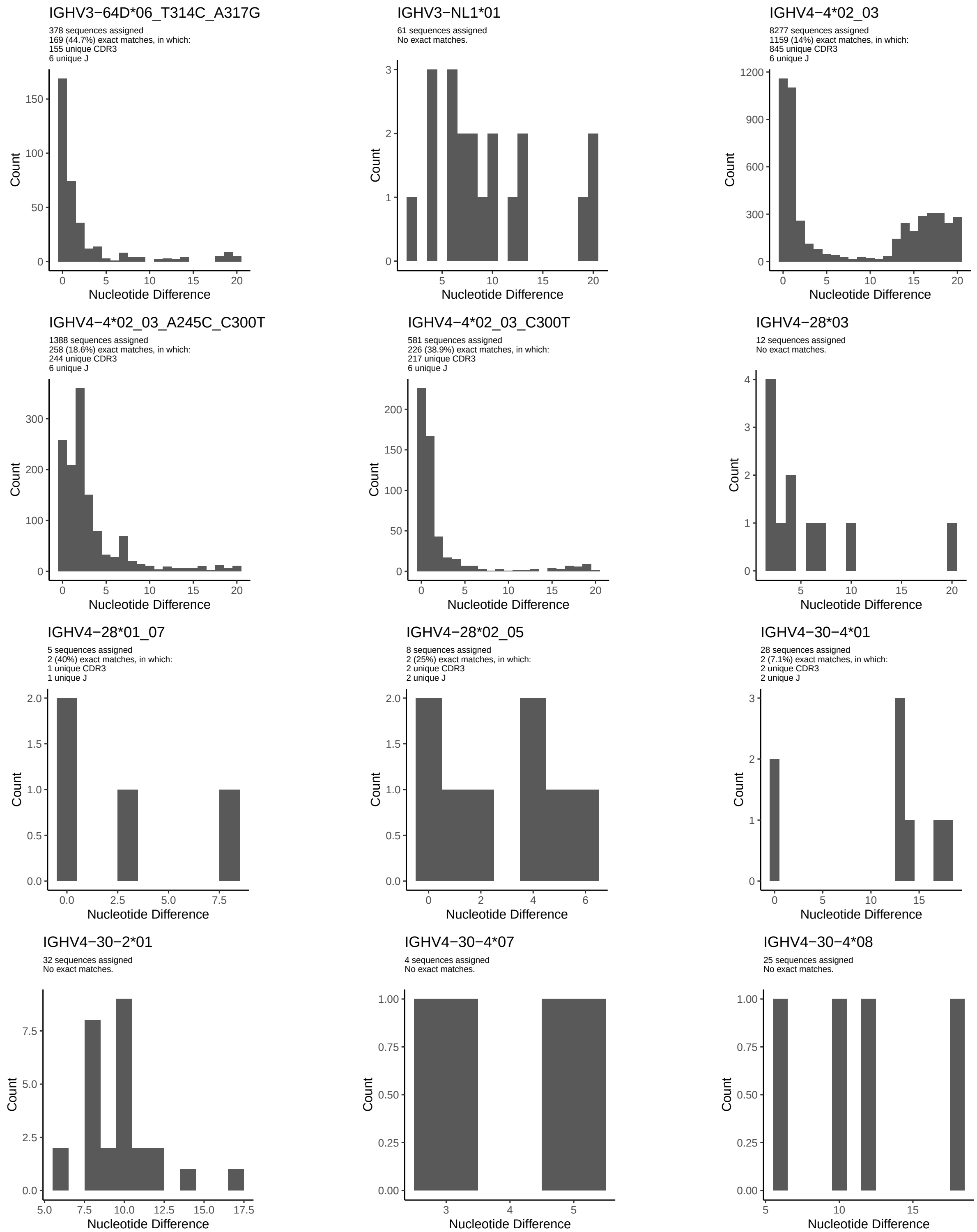


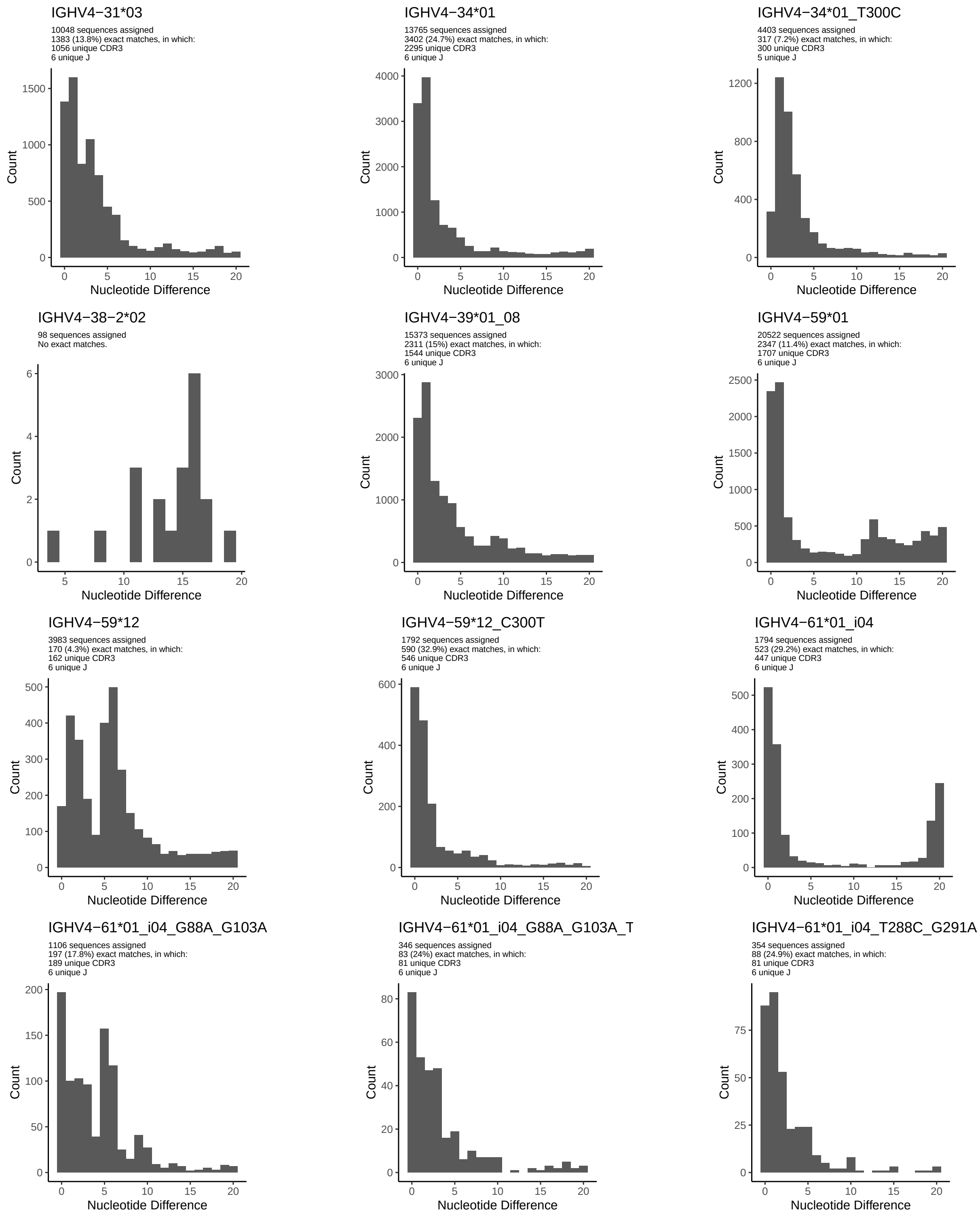
IGHV1−58*02
1412 sequences assigned
135 (9.6%) exact matches, in which:
105 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV1−58*01_03
1166 sequences assigned
192 (16.5%) exact matches, in which:
149 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV1−69−2*01
3 sequences assigned
No exact matches.
Count
Nucleotide Difference
IGHV1−69*01_12_13
7510 sequences assigned
1925 (25.6%) exact matches, in which:
1530 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV1−69*04_09
4692 sequences assigned
1190 (25.4%) exact matches, in which:
959 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV2−5*02
1071 sequences assigned
133 (12.4%) exact matches, in which:
94 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV2−26*01
461 sequences assigned
115 (24.9%) exact matches, in which:
86 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV2−70*01
642 sequences assigned
175 (27.3%) exact matches, in which:
126 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV2−70*15
886 sequences assigned
145 (16.4%) exact matches, in which:
111 unique CDR3
5 unique J
Count
Nucleotide Difference
IGHV2−70D*04
5 sequences assigned
No exact matches.
Count
Nucleotide Difference
IGHV3−7*01
3380 sequences assigned
388 (11.5%) exact matches, in which:
371 unique CDR3
6 unique J
Count
Nucleotide Difference
IGHV3−7*03
14885 sequences assigned
1287 (8.6%) exact matches, in which:
890 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV3−9*01
34 sequences assigned
No exact matches.
IGHV3−11*05
657 sequences assigned
75 (11.4%) exact matches, in which:
74 unique CDR3
6 unique J
IGHV3−11*06
5645 sequences assigned
712 (12.6%) exact matches, in which:
536 unique CDR3
6 unique J
IGHV3−13*05
558 sequences assigned
128 (22.9%) exact matches, in which:
95 unique CDR3
6 unique J
IGHV3−13*06
86 sequences assigned
No exact matches.
IGHV3−15*01
8621 sequences assigned
773 (9%) exact matches, in which:
523 unique CDR3
6 unique J
IGHV3−20*04
2467 sequences assigned
275 (11.1%) exact matches, in which:
188 unique CDR3
6 unique J
IGHV3−21*01
11426 sequences assigned
1966 (17.2%) exact matches, in which:
1359 unique CDR3
6 unique J
IGHV3−23*01_04
16777 sequences assigned
1833 (10.9%) exact matches, in which:
1162 unique CDR3
6 unique J
IGHV3−23*01_04_C288T_C300T_
1380 sequences assigned
341 (24.7%) exact matches, in which:
314 unique CDR3
6 unique J
IGHV3−23*01_04_C300T_A303G_
3982 sequences assigned
517 (13%) exact matches, in which:
481 unique CDR3
6 unique J
IGHV3−30−3*01
293 sequences assigned
8 (2.7%) exact matches, in which:
8 unique CDR3
3 unique J
Count
Nucleotide Difference

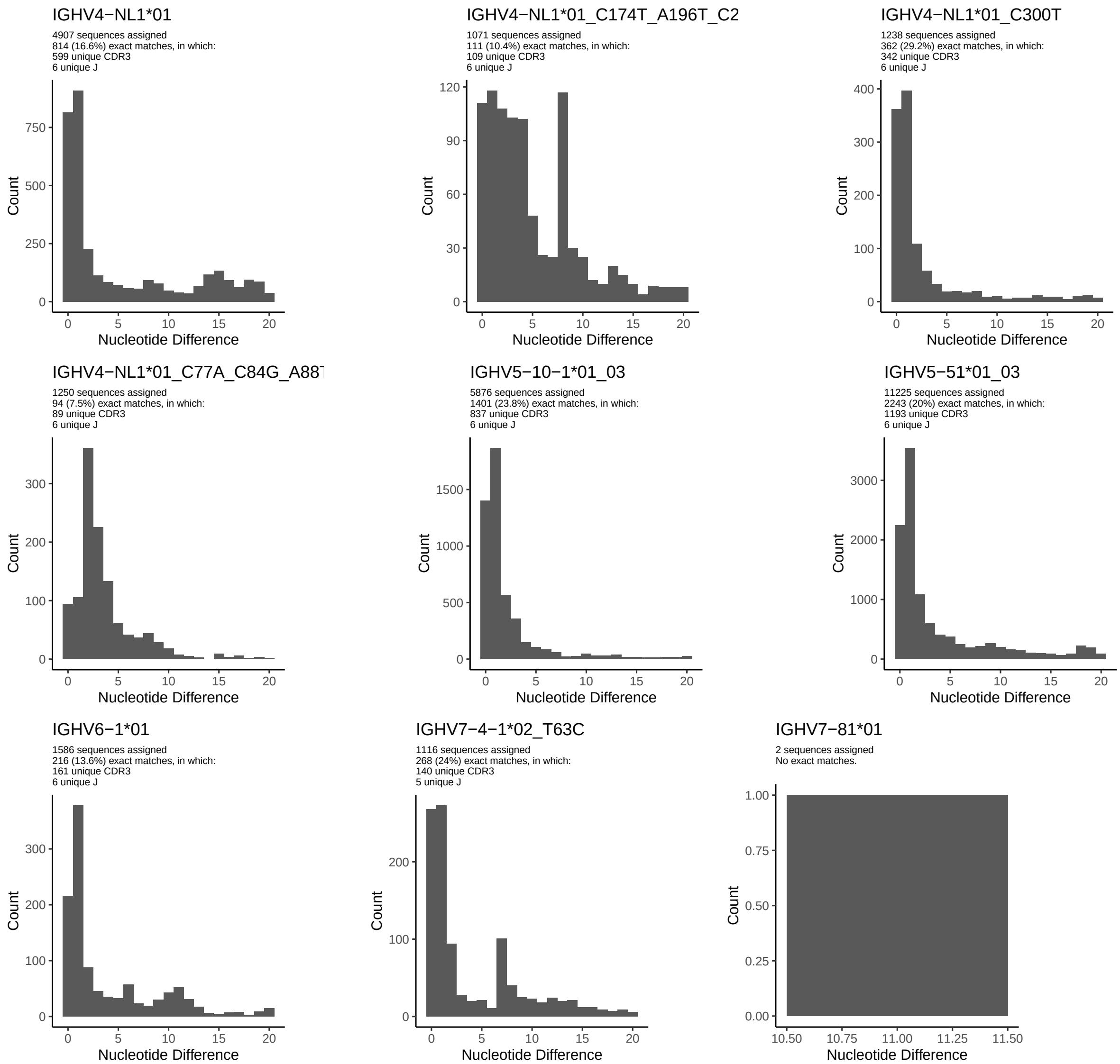

IGHV3−30*04
13042 sequences assigned
2931 (22.5%) exact matches, in which:
1997 unique CDR3
6 unique J
IGHV3−30*04_T288C
5381 sequences assigned
582 (10.8%) exact matches, in which:
534 unique CDR3
6 unique J
IGHV3−33*01
8361 sequences assigned
1231 (14.7%) exact matches, in which:
986 unique CDR3
6 unique J
IGHV3−33*08
12088 sequences assigned
1414 (11.7%) exact matches, in which:
1095 unique CDR3
6 unique J
IGHV3−35*01
1 sequences assigned
No exact matches.
IGHV3−43*01
2617 sequences assigned
110 (4.2%) exact matches, in which:
77 unique CDR3
5 unique J
IGHV3−48*01
11576 sequences assigned
1523 (13.2%) exact matches, in which:
1070 unique CDR3
6 unique J
IGHV3−49*03_05
1971 sequences assigned
399 (20.2%) exact matches, in which:
285 unique CDR3
6 unique J
IGHV3−53*05
28 sequences assigned
1 (3.6%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV3−53*01_02
5725 sequences assigned
169 (3%) exact matches, in which:
126 unique CDR3
6 unique J
IGHV3−64*01
343 sequences assigned
90 (26.2%) exact matches, in which:
65 unique CDR3
5 unique J
IGHV3−64*02_07
304 sequences assigned
13 (4.3%) exact matches, in which:
12 unique CDR3
4 unique J
Count
Nucleotide Difference


IGHV3−66*01
2271 sequences assigned
475 (20.9%) exact matches, in which:
364 unique CDR3
6 unique J
IGHV3−66*02
103 sequences assigned
No exact matches.
IGHV3−69−1*01
72 sequences assigned
No exact matches.
IGHV3−69−1*02
23 sequences assigned
No exact matches.
IGHV3−72*01
1588 sequences assigned
50 (3.1%) exact matches, in which:
36 unique CDR3
5 unique J
IGHV3−73*01_02
955 sequences assigned
242 (25.3%) exact matches, in which:
166 unique CDR3
5 unique J
IGHV3−74*01
19211 sequences assigned
398 (2.1%) exact matches, in which:
223 unique CDR3
6 unique J
IGHV3−74*01_A315G
3062 sequences assigned
202 (6.6%) exact matches, in which:
197 unique CDR3
6 unique J
IGHV3−74*01_T279C_A315G
7130 sequences assigned
98 (1.4%) exact matches, in which:
98 unique CDR3
6 unique J
IGHV3−43D*04_05
5 sequences assigned
No exact matches.
IGHV3−64D*06
2137 sequences assigned
394 (18.4%) exact matches, in which:
198 unique CDR3
6 unique J
IGHV3−64D*06_T267G_G275A_T2
948 sequences assigned
78 (8.2%) exact matches, in which:
76 unique CDR3
6 unique J
Count
Nucleotide Difference

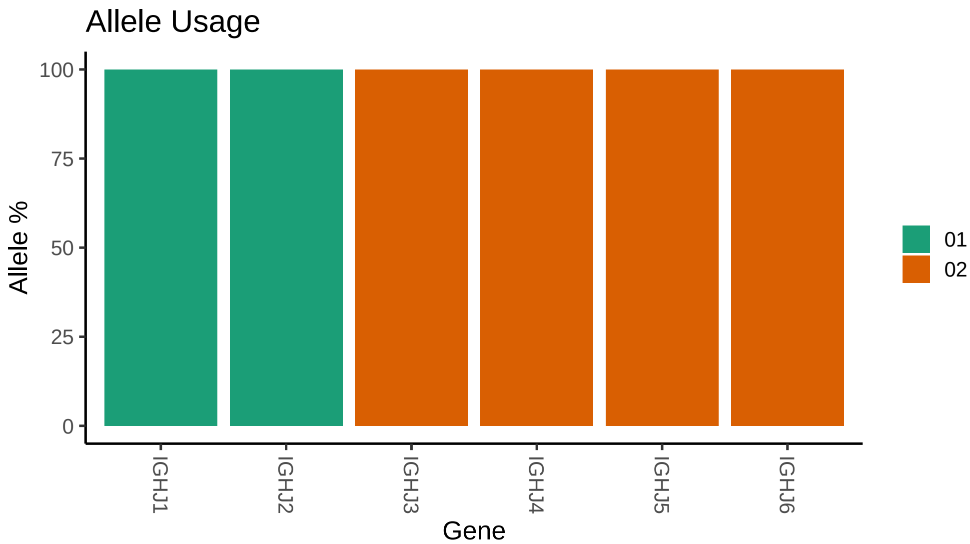

IGHV3−64D*06_T314C_A317G
378 sequences assigned
169 (44.7%) exact matches, in which:
155 unique CDR3
6 unique J
IGHV3−NL1*01
61 sequences assigned
No exact matches.
IGHV4−4*02_03
8277 sequences assigned
1159 (14%) exact matches, in which:
845 unique CDR3
6 unique J
IGHV4−4*02_03_A245C_C300T
1388 sequences assigned
258 (18.6%) exact matches, in which:
244 unique CDR3
6 unique J
IGHV4−4*02_03_C300T
581 sequences assigned
226 (38.9%) exact matches, in which:
217 unique CDR3
6 unique J
IGHV4−28*03
12 sequences assigned
No exact matches.
IGHV4−28*01_07
5 sequences assigned
2 (40%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV4−28*02_05
8 sequences assigned
2 (25%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV4−30−4*01
28 sequences assigned
2 (7.1%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV4−30−2*01
32 sequences assigned
No exact matches.
IGHV4−30−4*07
4 sequences assigned
No exact matches.
IGHV4−30−4*08
25 sequences assigned
No exact matches.
Count
Nucleotide Difference



IGHV4−31*03
10048 sequences assigned
1383 (13.8%) exact matches, in which:
1056 unique CDR3
6 unique J
IGHV4−34*01
13765 sequences assigned
3402 (24.7%) exact matches, in which:
2295 unique CDR3
6 unique J
IGHV4−34*01_T300C
4403 sequences assigned
317 (7.2%) exact matches, in which:
300 unique CDR3
5 unique J
IGHV4−38−2*02
98 sequences assigned
No exact matches.
IGHV4−39*01_08
15373 sequences assigned
2311 (15%) exact matches, in which:
1544 unique CDR3
6 unique J
IGHV4−59*01
20522 sequences assigned
2347 (11.4%) exact matches, in which:
1707 unique CDR3
6 unique J
IGHV4−59*12
3983 sequences assigned
170 (4.3%) exact matches, in which:
162 unique CDR3
6 unique J
IGHV4−59*12_C300T
1792 sequences assigned
590 (32.9%) exact matches, in which:
546 unique CDR3
6 unique J
IGHV4−61*01_i04
1794 sequences assigned
523 (29.2%) exact matches, in which:
447 unique CDR3
6 unique J
IGHV4−61*01_i04_G88A_G103A
1106 sequences assigned
197 (17.8%) exact matches, in which:
189 unique CDR3
6 unique J
IGHV4−61*01_i04_G88A_G103A_T
346 sequences assigned
83 (24%) exact matches, in which:
81 unique CDR3
6 unique J
IGHV4−61*01_i04_T288C_G291A
354 sequences assigned
88 (24.9%) exact matches, in which:
81 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV4−NL1*01
4907 sequences assigned
814 (16.6%) exact matches, in which:
599 unique CDR3
6 unique J
IGHV4−NL1*01_C174T_A196T_C2
1071 sequences assigned
111 (10.4%) exact matches, in which:
109 unique CDR3
6 unique J
IGHV4−NL1*01_C300T
1238 sequences assigned
362 (29.2%) exact matches, in which:
342 unique CDR3
6 unique J
IGHV4−NL1*01_C77A_C84G_A88
1250 sequences assigned
94 (7.5%) exact matches, in which:
89 unique CDR3
6 unique J
IGHV5−10−1*01_03
5876 sequences assigned
1401 (23.8%) exact matches, in which:
837 unique CDR3
6 unique J
IGHV5−51*01_03
11225 sequences assigned
2243 (20%) exact matches, in which:
1193 unique CDR3
6 unique J
IGHV6−1*01
1586 sequences assigned
216 (13.6%) exact matches, in which:
161 unique CDR3
6 unique J
IGHV7−4−1*02_T63C
1116 sequences assigned
268 (24%) exact matches, in which:
140 unique CDR3
5 unique J
IGHV7−81*01
2 sequences assigned
No exact matches.
Count
Nucleotide Difference

3 Allele usage in potential haplotype anchor genes



4 Haplotype plots

5 Configuration settings

```
## Repertoire file: /misc/work/jenkins/PRJNA491287/M64 075/M64 075/M64 075/M64 075/cd/84fb5ad24d6b129b5
##
## Germline reference file: /misc/work/jenkins/PRJNA491287/M64 075/M64 075/M64 075/M64 075/cd/84fb5ad24
##
## Novel allele file: /misc/work/jenkins/PRJNA491287/M64 075/M64 075/M64 075/M64 075/cd/84fb5ad24d6b129
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 24 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 24 01_A315G_C317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 46 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 46 01_C104T_G107A_T112C_G126A_C171A_C173A_A175G_G176A_G188T_G19
##
##
## Unexpected N in novel sequence IGHV1 46 01_C104T_G107A_T112C_G126A_C171A_C173A_A175G_G176A_G188T_G19
##
##
## Unexpected N in reference sequence IGHV3 23 01_04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C300T_A303G_A317G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 23 01_04_C288T_C300T_A303G_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 04_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 64D 06: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 64D 06_T314C_A317G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 64D 06_T267G_G275A_T279C_T314C_A317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 74 01: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV3 74 01_A315G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_T279C_A315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 4 02_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 12: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 59 12_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 61 01_i04: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_T288C_G291A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_G103A: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 61 01_i04_G88A_G103A_T288C_G291A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 NL1 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C300T: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C77A_C84G_A88T_A106G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 39 01_08: replacing with gap
```

```
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C174T_A196T_C207G_A237C_C300T: replacing with gap
```